

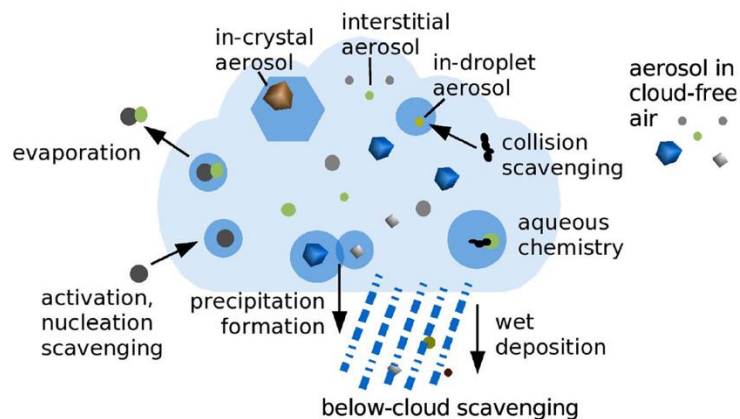
MAM4xx: A Performance-Portable Aerosol Model for Next-Generation Earth System Simulations

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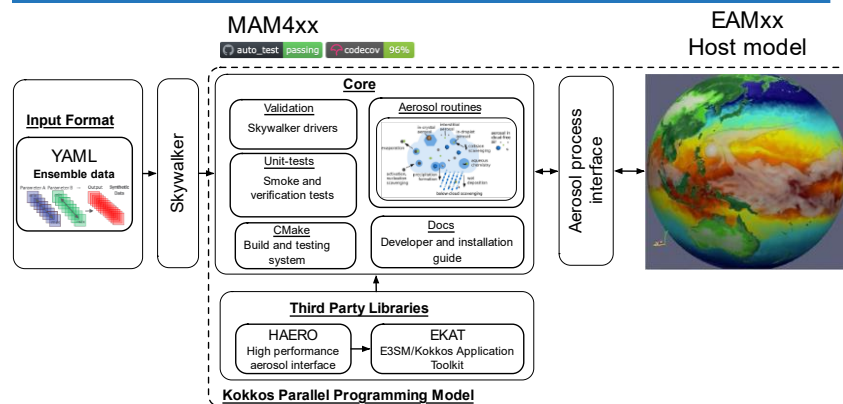
What is MAM4xx?

Prognostic Aerosols in EAMxx



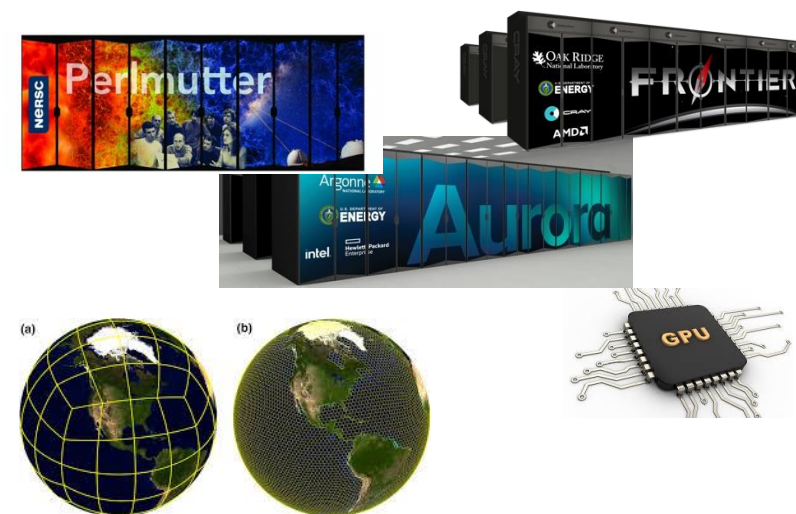
- ✓ Detailed prognostic aerosol interactions
- ✓ Builds upon the Modal Aerosol Model (MAM4)

Performance-Portable Architecture



- ✓ Modern C++/Kokkos for portability
- ✓ Designed for next-generation HPC systems

High-Resolution Modeling at Exascale



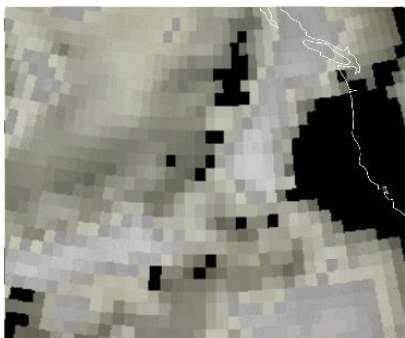
- ✓ Optimized for GPU platforms
- ✓ Enables higher-resolution simulations
- ✓ Science at scale on exascale architectures

*EAMxx: E3SM Atmospheric Model
written in C++

Legacy Fortran to Modern C++/Kokkos

- ❑ Modern C++
- ❑ Kokkos portability
- ❑ Cmake
- ❑ Automated testing
- ❑ Code coverage analysis

**MAM4
In
EAM**



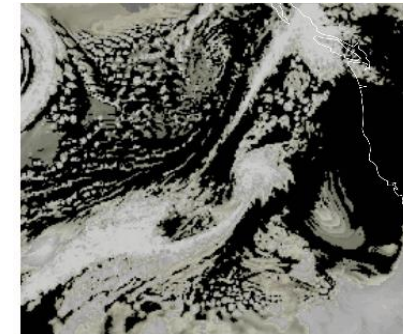
$\Delta x = 100 \text{ km}$

Refactor Fortran code

Port to C++/Kokkos

Integrate to EAMxx

**MAM4xx
In
EAMxx**



$\Delta x = 3 \text{ km}$



F >> **C++**



Fortran

C++/Kokkos

Refactoring Fortran

Before

 Long Subroutines

 Mixed logic flow

 Hard to reuse

 Dead/untested code present

After

 Multiple very short subroutines

 (Ideally) Each routine does one task

 Reusable and maintainable

 100K+ lines removed

Philosophy

- ✓ “True” refactor and port — *identify, don’t fix bugs*
- ✓ Remove dead and non-MAM4 code
- ✓ Keep results **BFB** (bit-for-bit identical)
- ✓ Driven by **tests** — end-to-end validation

Validation Framework for the C++ Port

**E3SMv2
(Fortran)**

**Extract
Column Data**

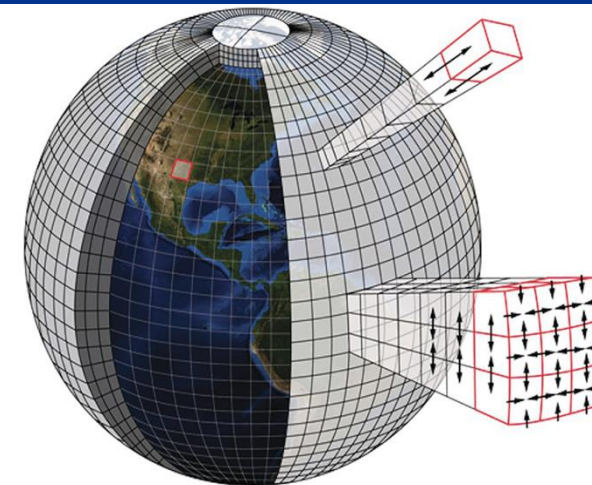
**Skywalker
Format**

**C++ Routine
Tests**

New infrastructure to generate and validate test data for the C++ port.

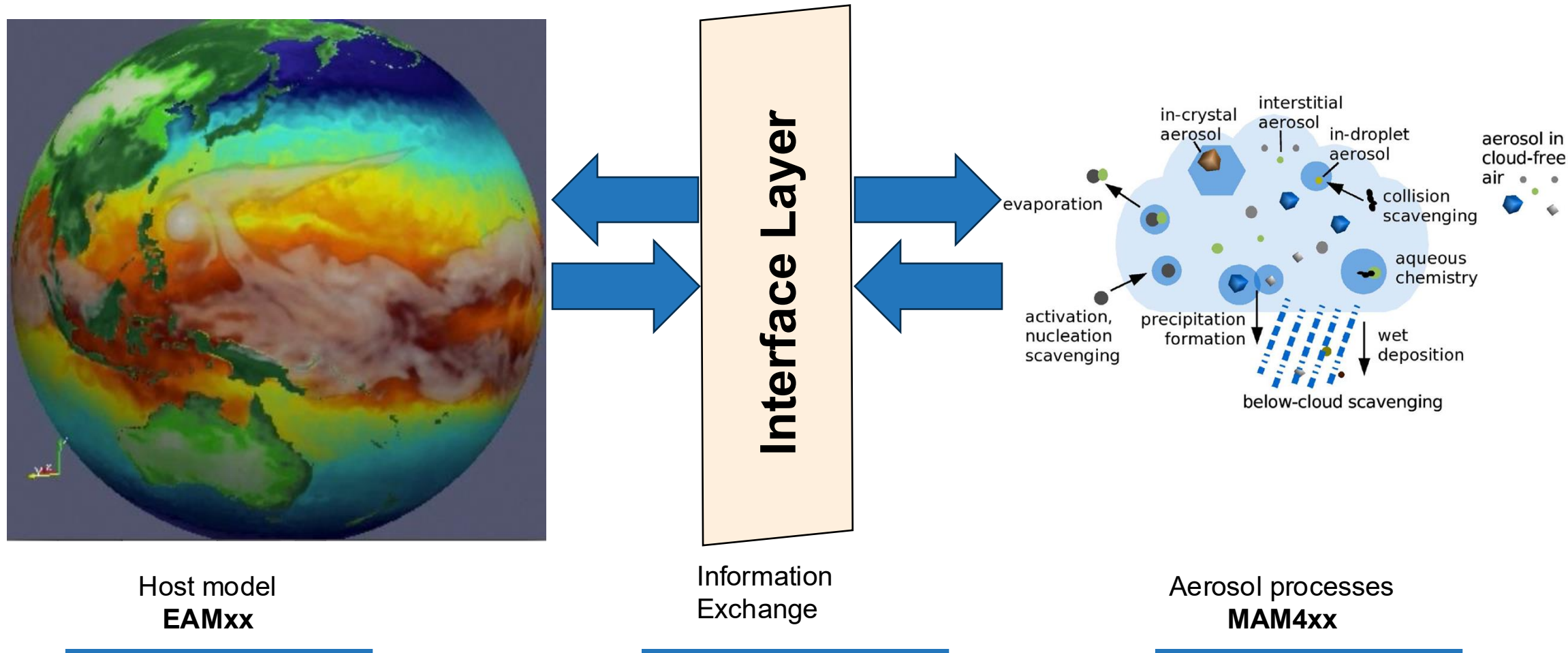
Key Features

- ✓ Flexible and extensible infrastructure
- ✓ Extracts **massive amounts of data** efficiently
- ✓ Supports **any column, time step, or vertical level**
- ✓ Integrates with **automated code generation**
- ✓ Simplifies regression and unit testing of ported routines



EAM column/grid validation data

EAMxx-MAM4xx Integration

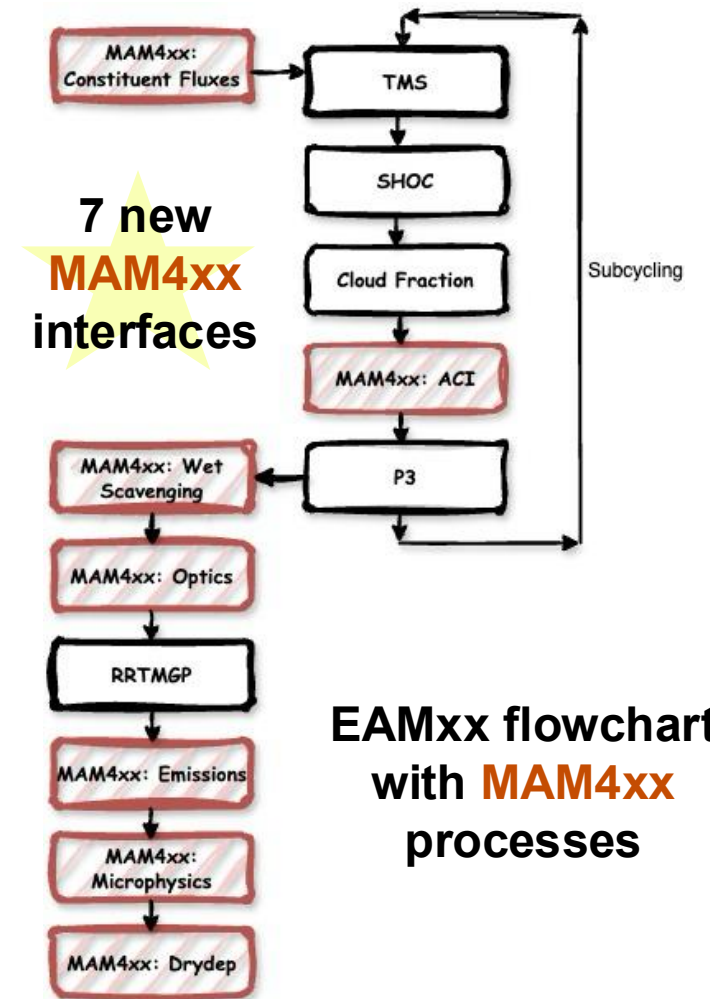


MAM4xx: Simulation Framework

- Supports a wide range of horizontal resolutions on **GPUs**
✓ **ne4** (~800 km) to **ne1024** (~3 km)
- Fully supported on **all DOE leadership class machines**
- **CI/CD**: Nightly tests on all E3SM-supported and DOE machines (**CPUs** and **GPUs**)



Easy-to-use compset!
F2010-EAMxx-MAM4xx



Challenges

New host model (EAMxx)

Limited I/O support

Physics is untested at high resolution

Numerical instabilities at fine grid scales

Race conditions on GPUs

Variability across GPU machines

Long simulation crashes

What helped

Modular design to integrate incrementally

Extended existing capability-modified metadata

Hierarchical testing (unit → regression → BFB)
at low to high resolutions

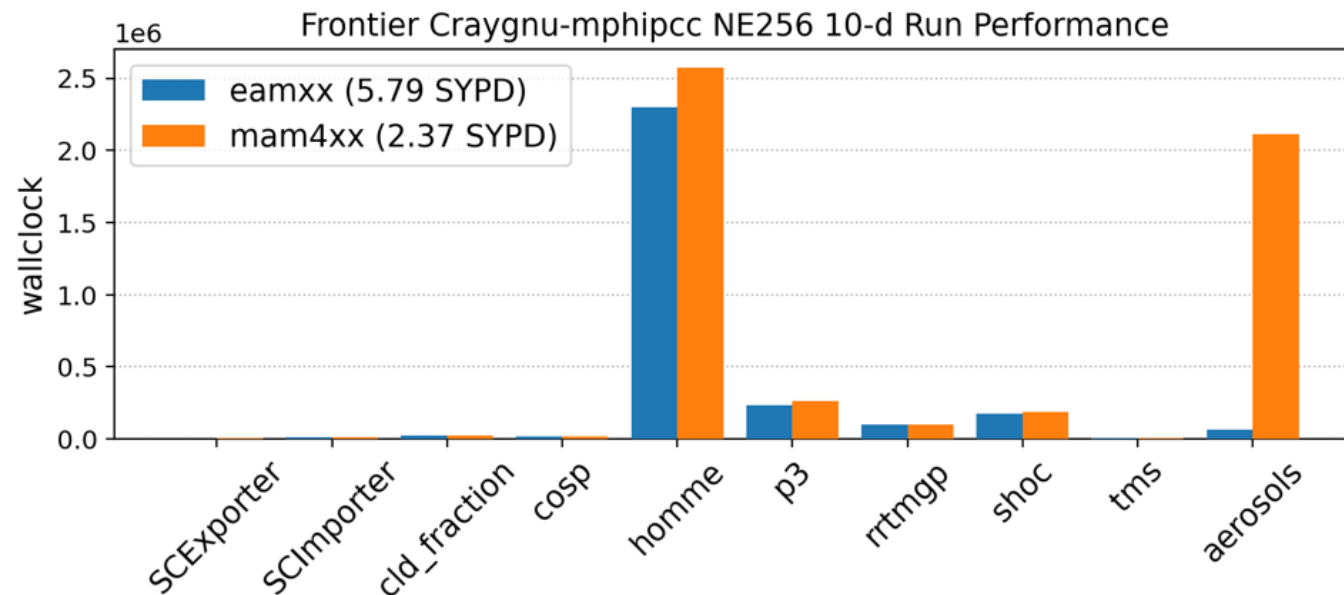
High-resolution testing on target machines

Code reviews, compiler sanitizers, vendor tools (e.g.,
CUDA-MEMCHECK)

Collaboration with machine POCs and vendors

Continuous testing, nightly runs, reproducibility
checks

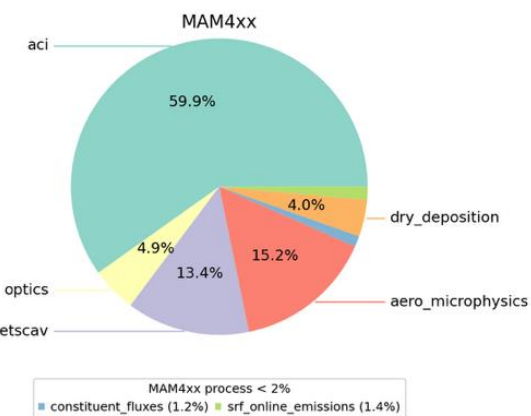
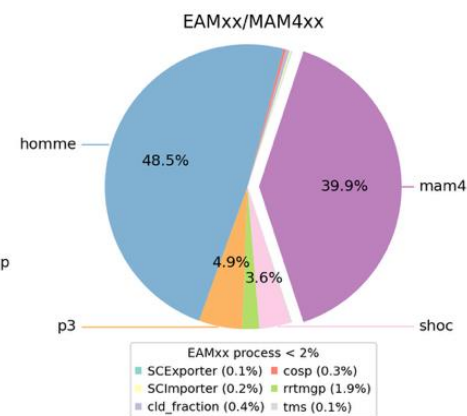
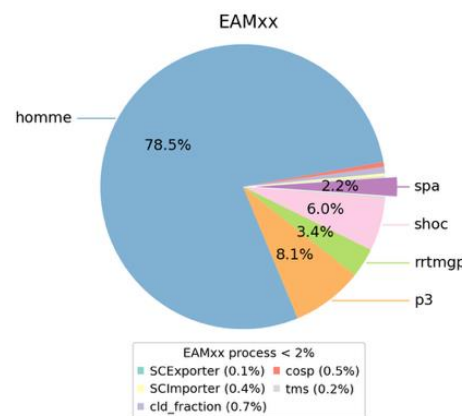
Performance at ne256pg2 (~12km)



- ~2.4× cost of **EAMxx** at ne256pg2 (SYPD)
- **+31 prognostic species**
- Most expensive parameterized physics
- Slightly cheaper than HOMME dynamics

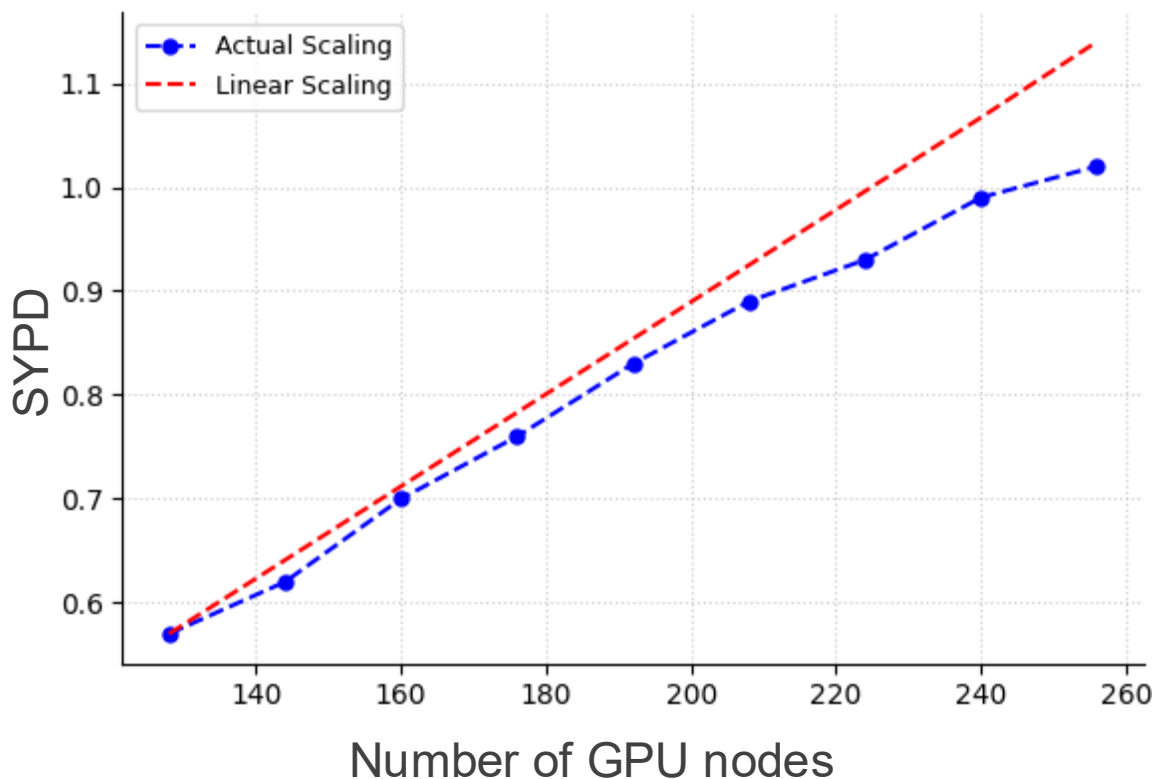
Cost Distribution:

- **MAM4xx**: ~40% of total cost
- **HOMME**: ~49% of total cost
- **ACI** is the most expensive process within **MAM4xx**

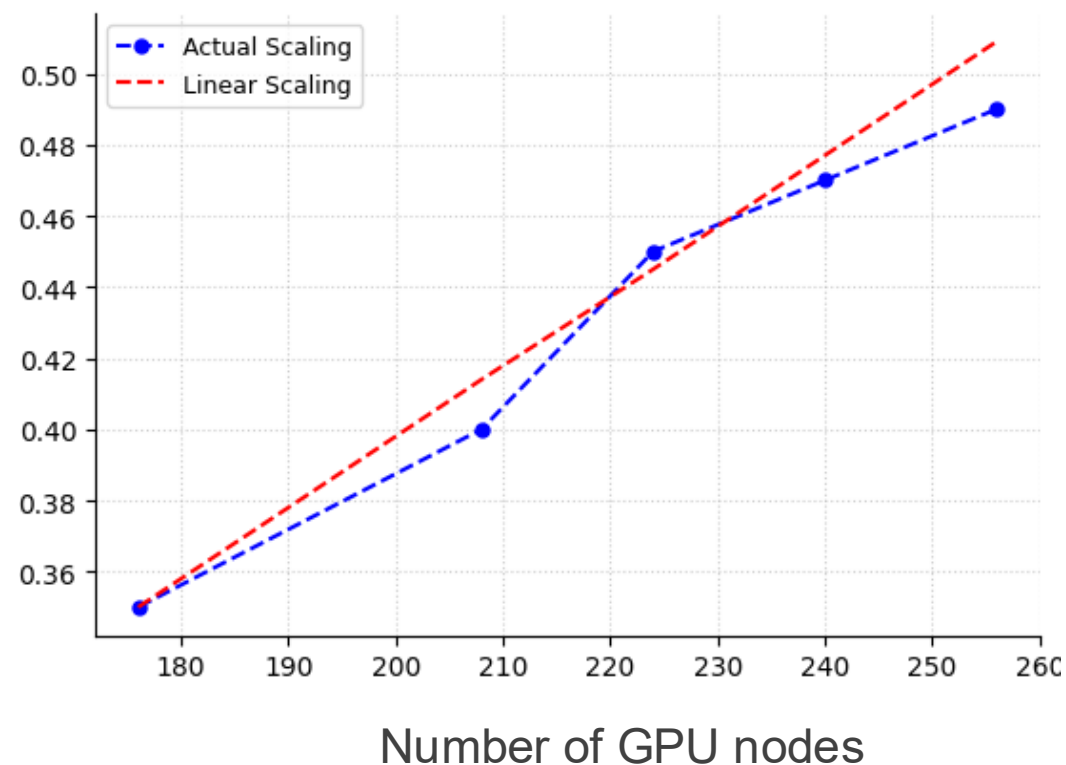


MAM4xx: Strong Scaling

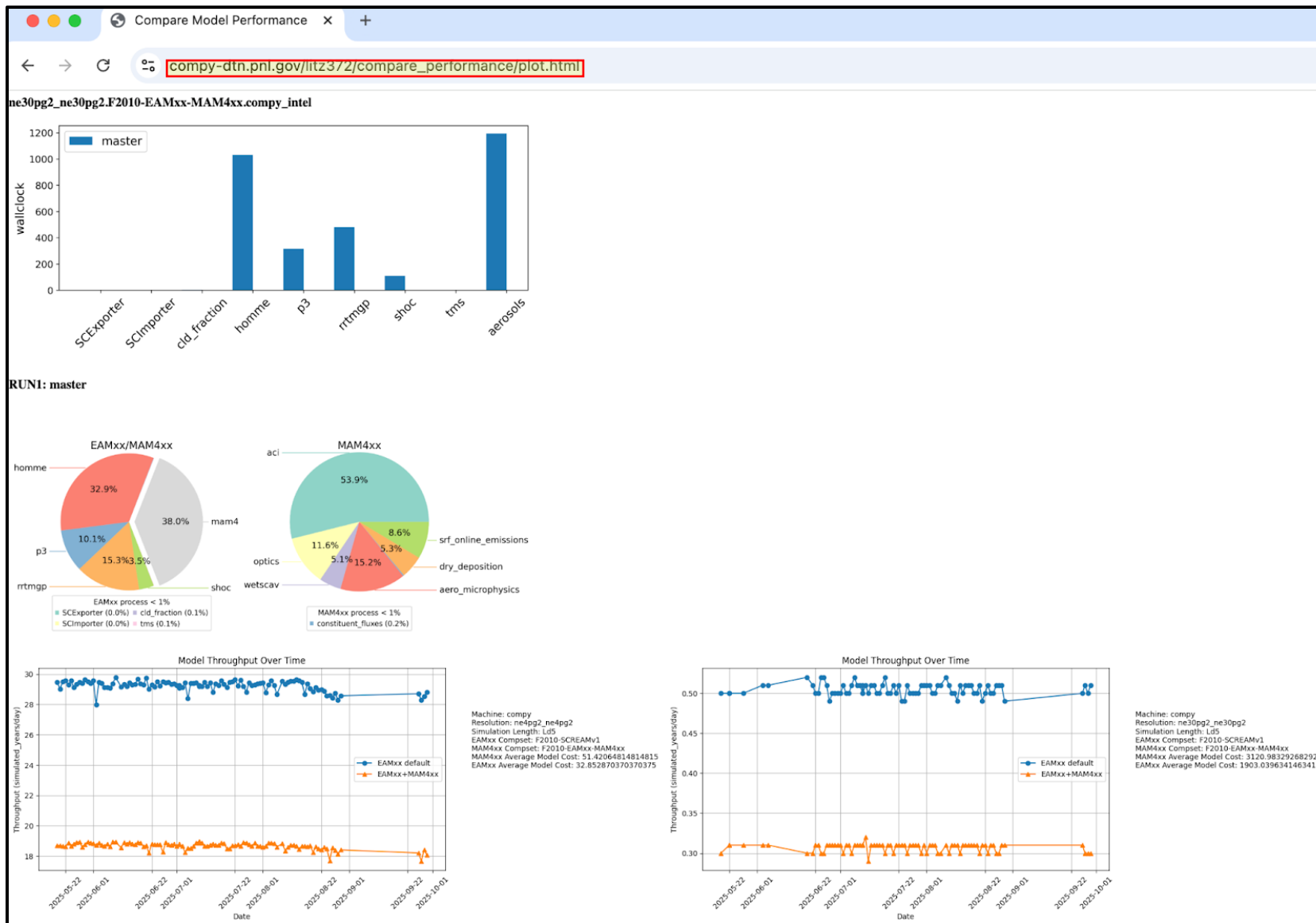
PM-GPU at ne256



Frontier at ne256



*MAM4xx cost distribution remains consistent as GPU nodes increase (not shown).



Compy:

https://compy-dtn.pnl.gov/litz372/compare_performance/plot.html

PM-GPU:

https://portal.nersc.gov/cfs/e3sm/litz372/performance_data/performance_comp_ne30pg2_ne30pg2_Ld5.png

Summary and Next steps

- The EAMxx-MAM4xx simulation framework is fully functional
- Stable simulations at target resolution (ne256pg2) for 1+ year
- Runs successfully on all GPU and CPU machines (LCFs and E3SM)
- Nightly tests are run on both GPU and CPU architectures
- Performance: **~2.4x slower** than EAMxx-SPA at the target resolution
 - Scales well for strong scaling
 - Resolution sensitivity– needs more analysis
- **Next steps:**
 - Conduct a deep performance analysis to identify optimizations:
 - Revise complex kernels, serial loops, and super cycling
 - CI/CD Performance dashboard
 - Add new features as required

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